

Immerse Education AI

Stocks Prediction

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31/07/2025

Abstract

This report is about teaching artificial intelligence to learn about the stock market. To feed AI old stock market data are used. There are a lot of parameters which affect the market which resulted in it having some loss. Even though it had that loss there were ways to improve it and increase its accuracy. We tried to change everything and always compare them to find which would be the best to make it the most accurate it can possibly be. The problem was sometimes people had a hard time trying to analyze these types of big graphs which could be affected by a lot of things and we tried to solve it by giving the data to artificial intelligence and making it learn so it could help us when it learned enough.

1. Introduction

1.1 Background

Artificial intelligence is a simulation of human brains that can learn something new and some abilities such as comparison, making decisions and problem solving [1]. Machine learning is the learning ability of the AI which is autonomous and in really fast ways of a big dataset [2]. We used regression graphs because we wanted it to calculate how much it will increase or decrease instead of making something simple like calculating the probability of its increasing or decreasing. Regression graphs learn the data and take the average of it and guess based on the average while classification such as just guessing if it will go up or down learns how the data is separated and separates them by where it finds its most type [3]. Deep learning is a way of machine learning but the difference is it uses neural links and has a better type of understanding [4]. LSTM is Long Short-Term Memory and it is a type of deep learning. It is used for some

time series problems like the stocks, weather or the updated data of something which is generally prices or values. LSTM can remember patterns and the relationship between some parameters. It is mostly used in analysis. LSTM is also a better choice next to simple RNN's because it can remember more things and can learn patterns easier which makes it a more complicated thing [5-6].

1.2 Objectives and Aim

The objectives are teaching artificial intelligence about graphs and data about numbers. We hope that artificial intelligence can learn and make meaning of the data that it is given which is also used for calculating the predicted price. Making it as most accurate as we can is also the biggest objective. The aim of this project was firstly guessing how a stock will change such as increasing or decreasing. As the time passed by it evolved to guessing the amount of the increase or the decrease. The main purpose is actually making an accurate bot that can predict the future of the stock market faster and better than humans which would also preserve the effort that is given on that.

2. Datasets

The most important and the basic dataset that we have used is yahoo finance which was needed to find the old stock prices in numbers. We preferred yahoo finance because it was the most reliable we found and also it was simple to use as the library. We took the data interval as 1 day for the whole project but changed the starting point of it to find the most accurate one that we can use because sometimes the market crashes out or there are some special times that the market prices goes really unexpected so we needed to take as many information as we can but at the

same time the most predictable we can. First we tried taking information starting from 2000/01/01 to 2025/01/01. But it wasn't successful enough because it was a really long part so we changed the starting point to 2010/01/01 which ended up still not successful enough but better than the 2000/01/01. So we changed to go closer for better results and at the end we chose 2020/01/01 to 2025/01/01 which was the most accurate that we have found. We also took the information of some macro economic variables which affected the prices a lot. They are VIX, DYX, US10Y, XAUUSD, SPX, DJI, IXIC, RUT. We chose head, close, high, low and the volume data [7].

3. Methodology

3.1 Tools Used

We used some libraries while doing the project and also the coding language Python. The libraries that we used can be listed as pandas, numpy, keras, Tensorflow, matplotlib, yfinance, sklearn. All of these libraries were necessary while doing the project. Some of them were about the graphs, some of them were about the artificial intelligence model and some about the maths and finance to take the data from the internet which is also mentioned at the dataset part because it was the most important one so we can also take today's data too.

3.2 Data Preparation

We used AAPL, NVDA and TSLA stocks data because they were the most reliable ones on the market because they have existed for a long time and they don't generally have big movements on their prices except crisis which all of them have. We also used the interval between 2020/01/01 - 2025/01/01. We matched the macro economics data and the technical data in the

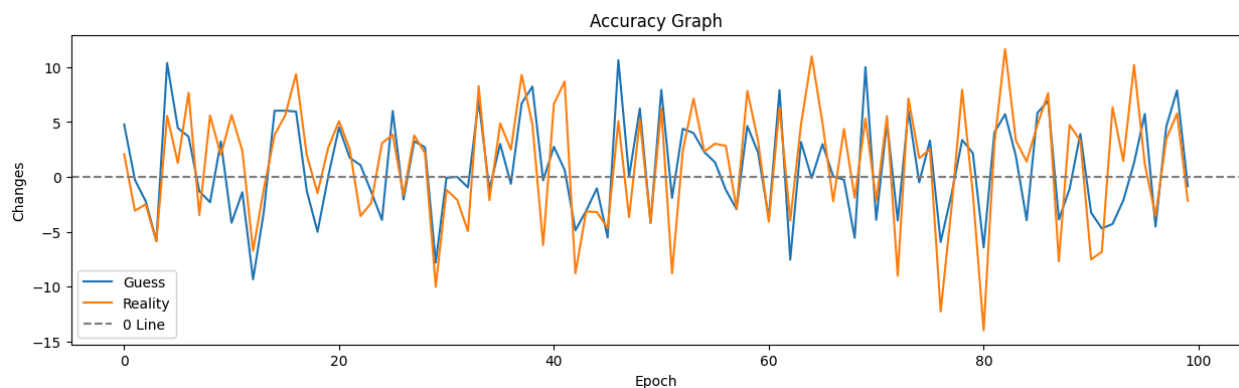
same group so we could use it in one piece. We also did not forget to normalize the data so the AI could understand it and convert it back so people could be able to understand. We have chosen 0.2 as the train test split because it worked the best as we experienced.

3.3 Model Implementation

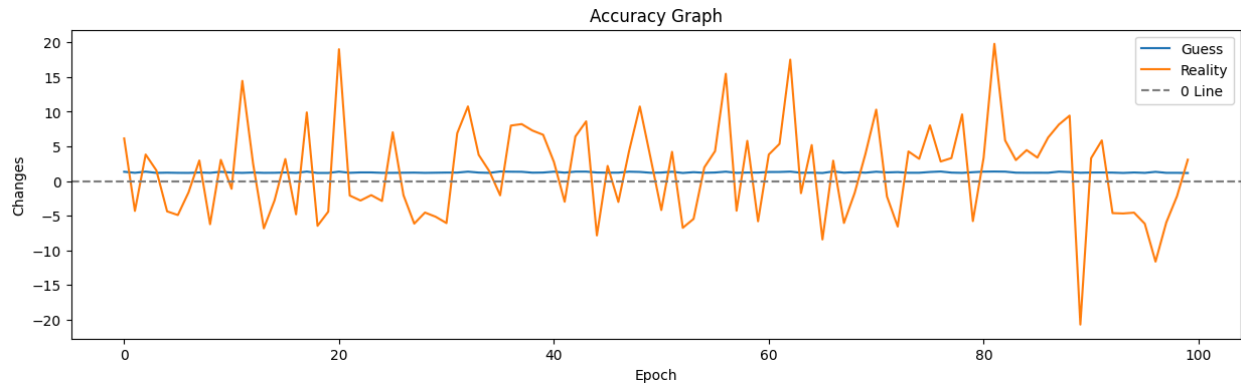
We used LSTM as our model. We preferred using 2 hidden layers to get the best results and the first one had 128 neurons, the second one had 64 neurons. It did not have any dropouts because they made the learning slower and our AI was underfitting so it was useless from our point of view. We did not use any activation with the dense because it was doing regulation. The model is compiled by mse because by considering our observations mse had better results compared to mae.

4. Results

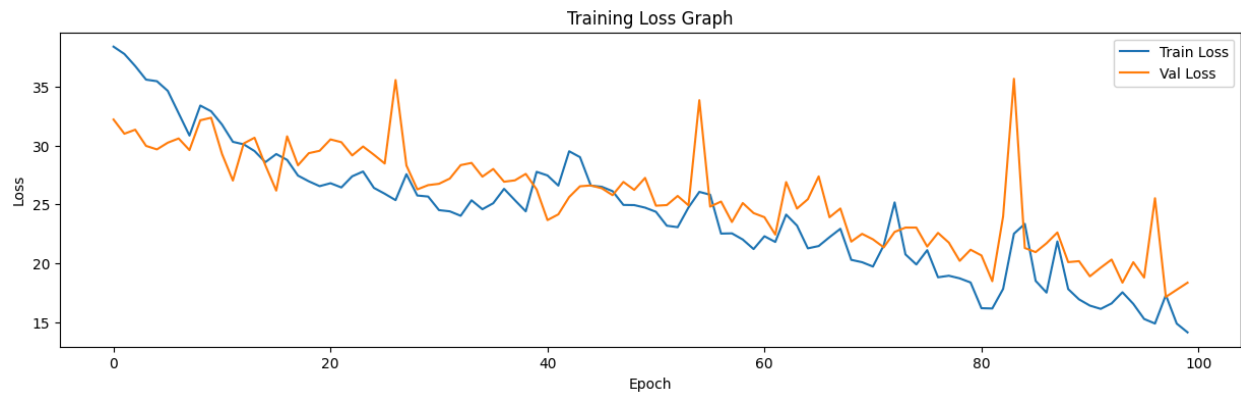
LSTM:



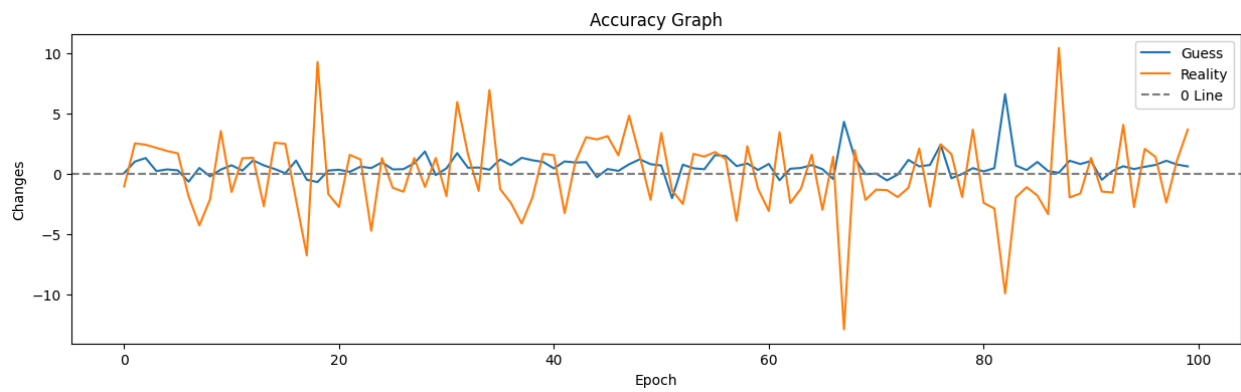
RNN:



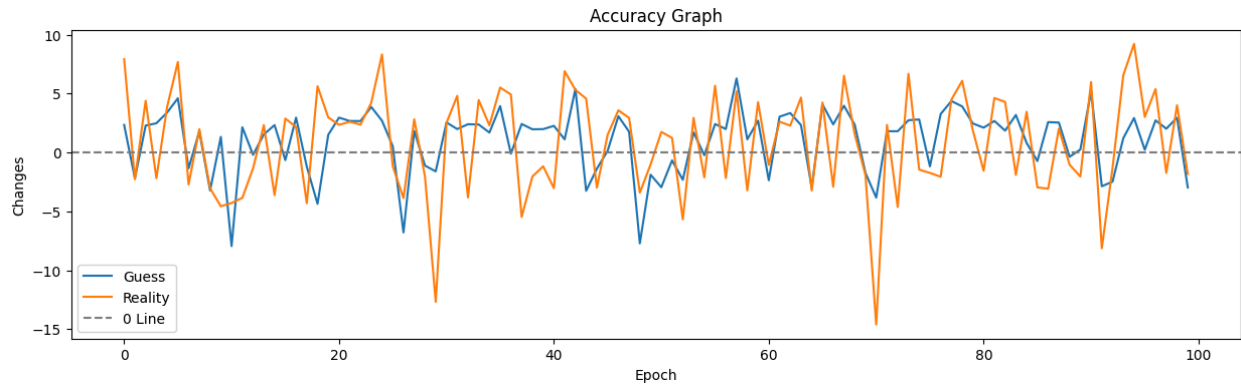
200 Epochs:



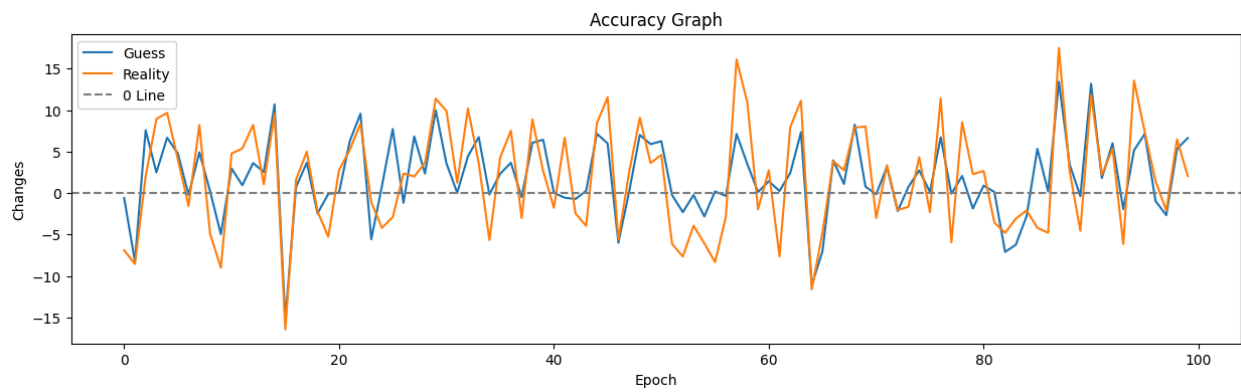
Guessing 1 days later:



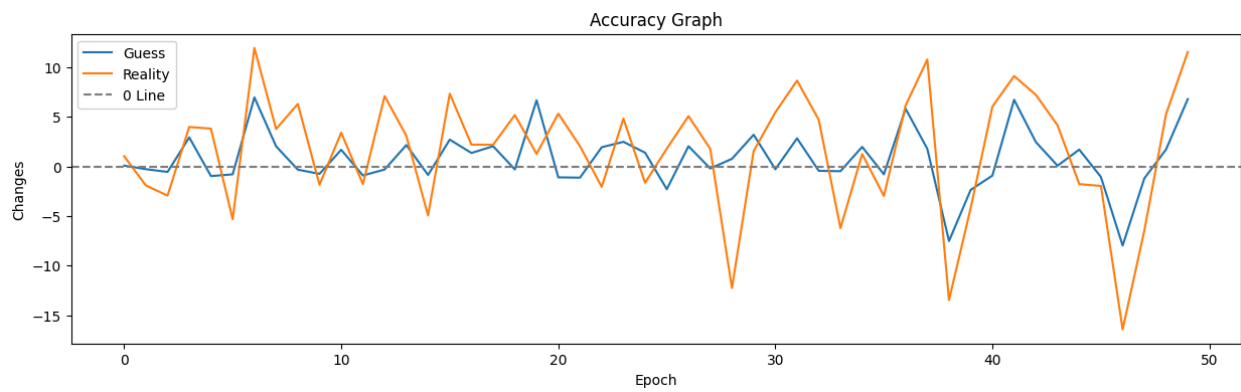
Guessing 5 days later:



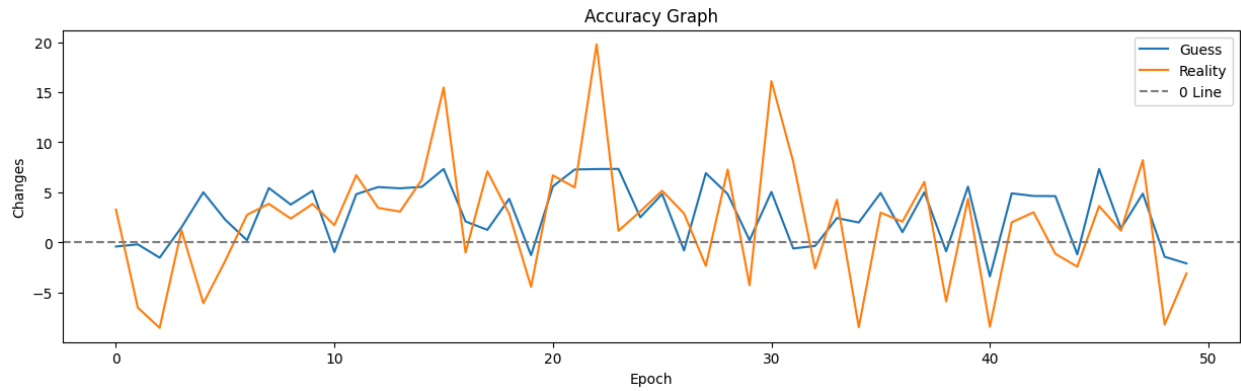
Guessing 10 days later:



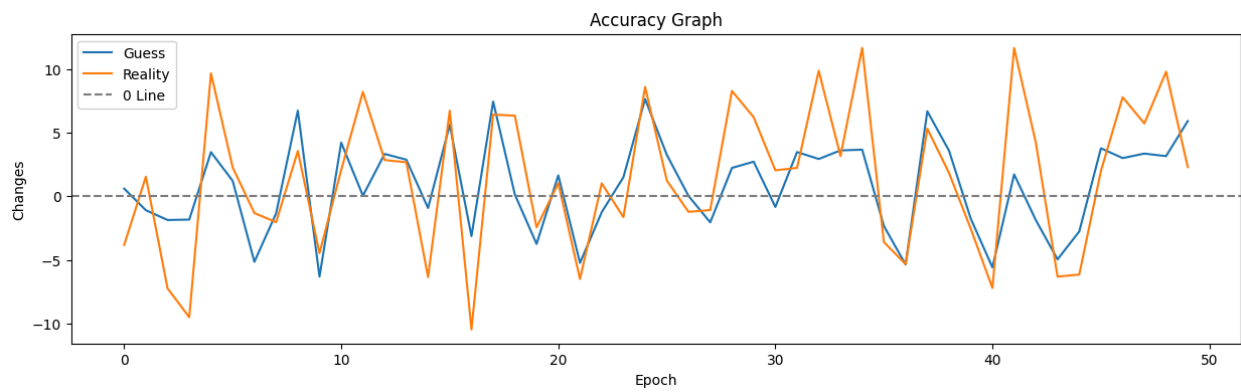
Looking for 14 days back:



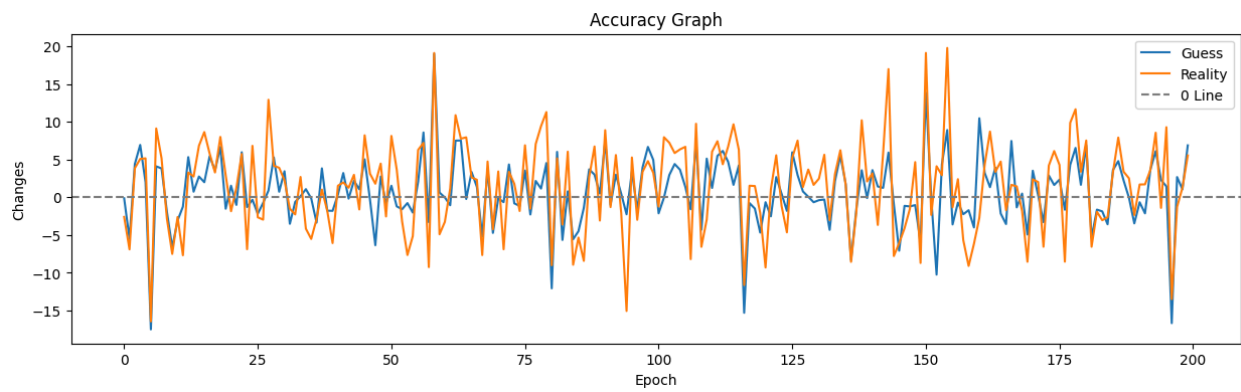
Looking for 30 days back:



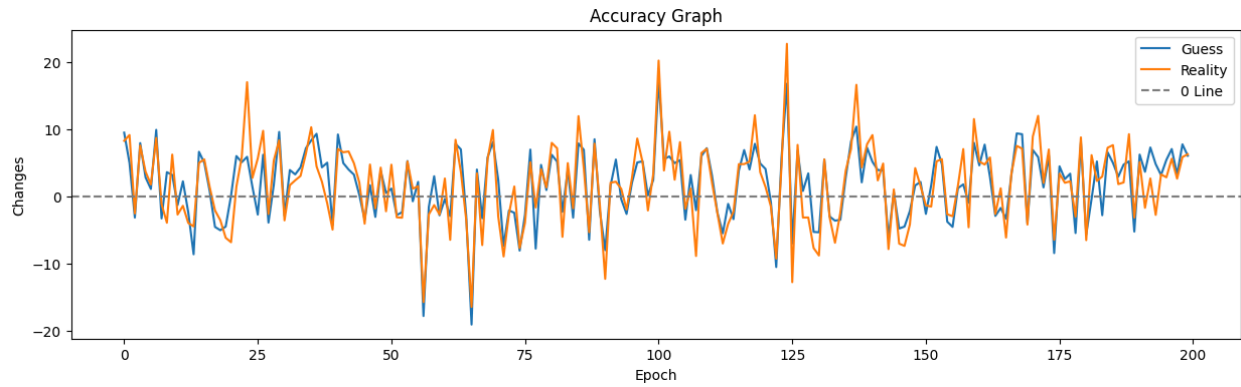
Looking for 60 days back:



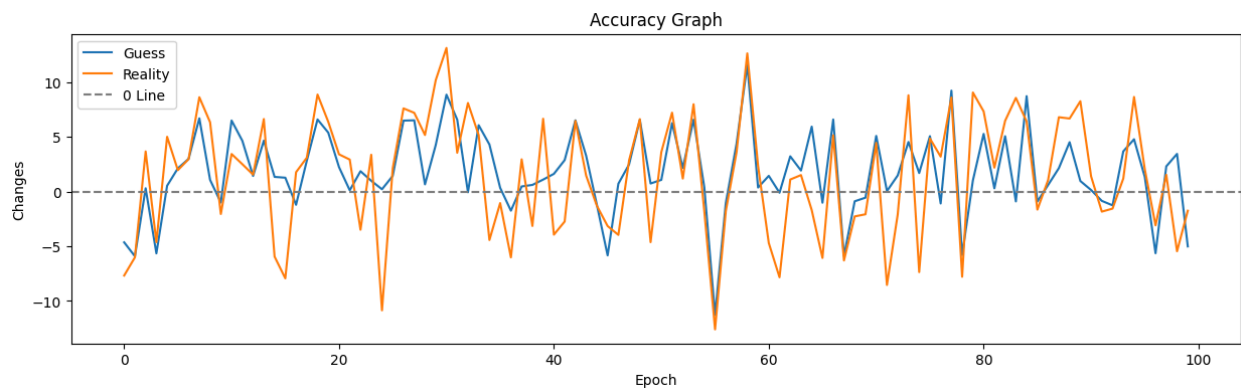
Without macro economics data:



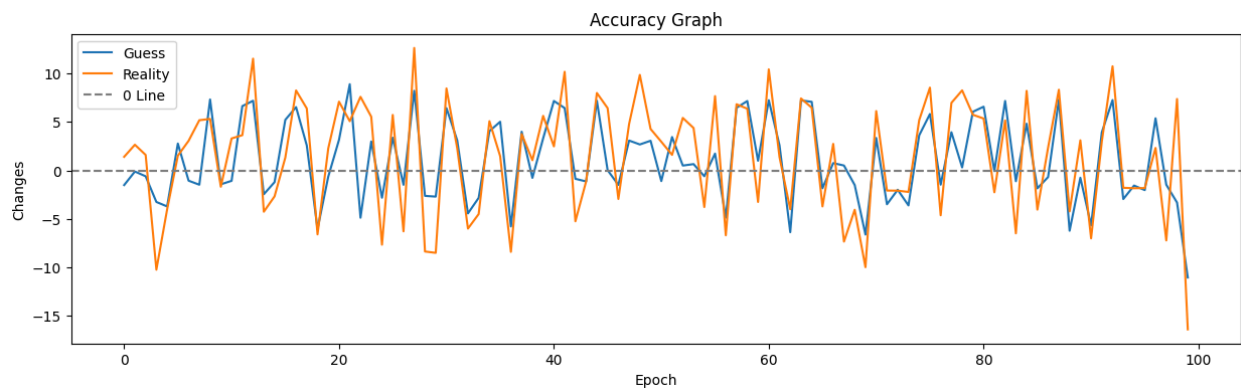
With macro economics data:



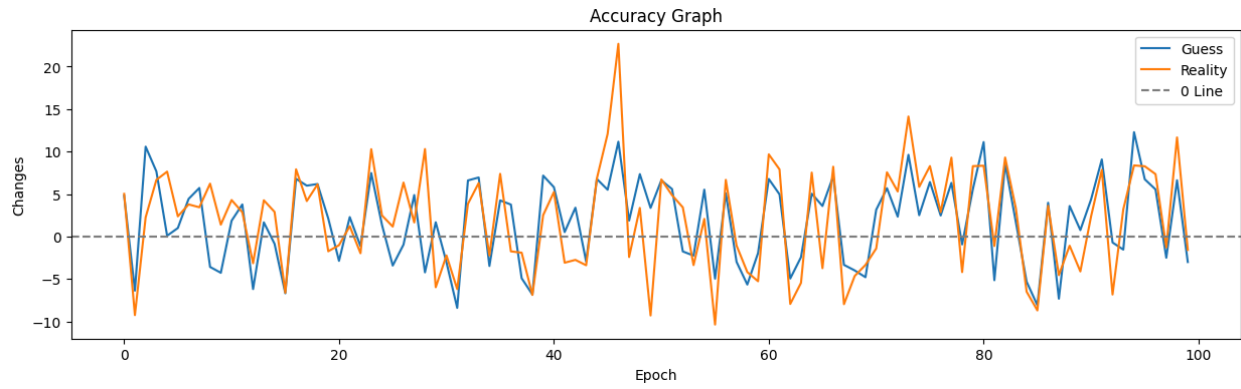
Layers as 64:



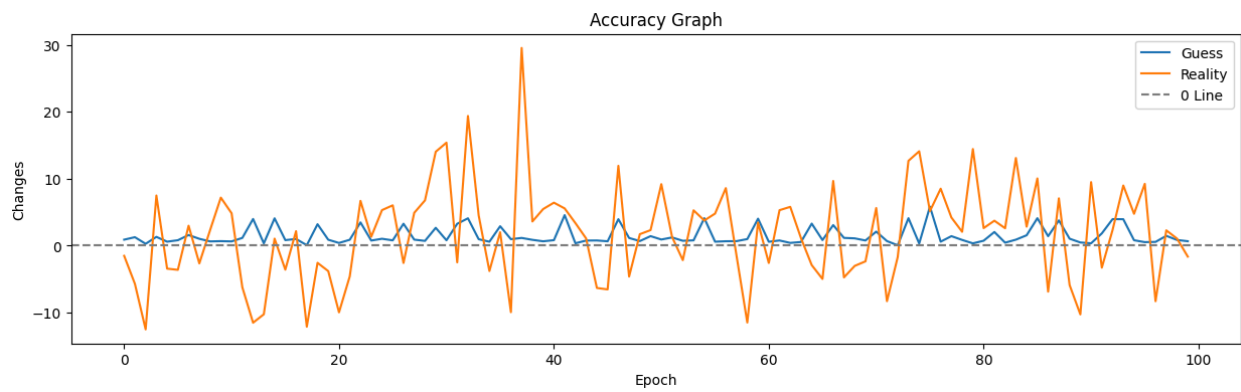
Layers as 64 and 32:



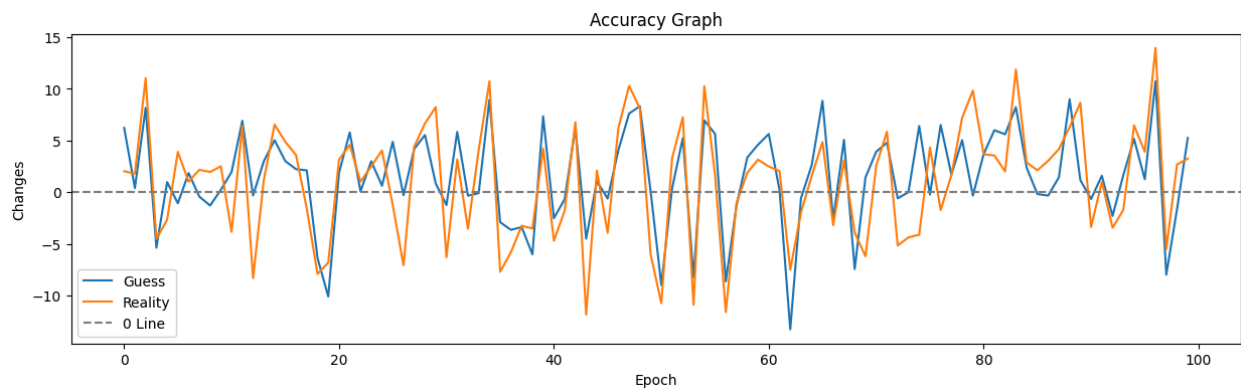
Layers as 128 and 64:



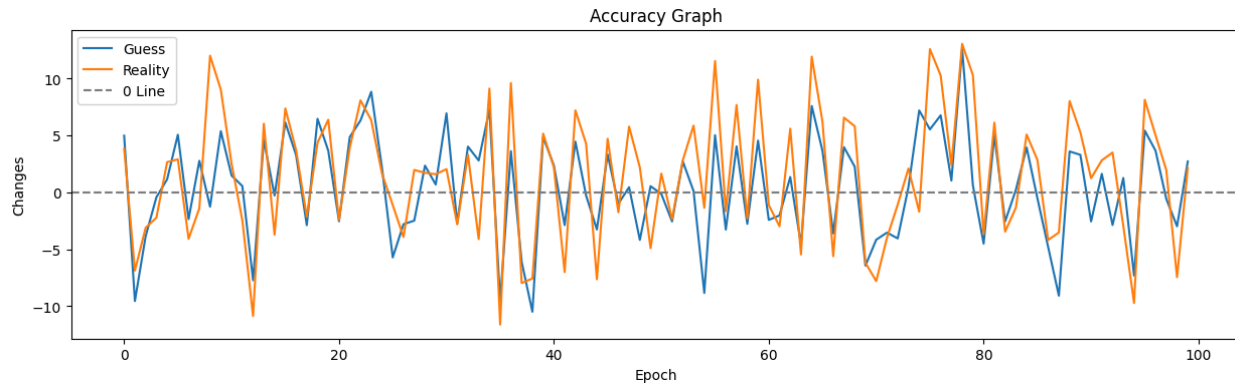
2000 - 2025:



2010 - 2025:



2020 - 2025:



5. Discussion

We can obviously see that LSTM is much better than RNN for our problem because we need a lot of variables so it needs more memory which is a great problem of RNN because RNN can forget some information [8]. We also tried different amounts of epochs to see if the loss continues decreasing to make it its best version. One thing which we tried is also the the day number that it will guess about the feature and we found out that the best number for it was 10 out 1, 5 and 10 because it needed to be a short time period to not have a crisis between the date of the guess and the reality of the markets price and at the same time more meaningful than 1 because there can be fluctuation in a day. About the time that it checked it shouldn't be too long to not confuse it with unnecessary information but at the same time as long as it could understand the momentums and the general situation of the stock. Also the extra macro economics information helped it to increase accuracy by a visible difference because they also affect the stock market and without them it would be missing something so AI would be less successful at understanding the patterns. The number of hidden layers and neurons is the second most important thing that affects accuracy because it shows how intelligent the computer is and how it can calculate the change. The data is also a lot important because it is the quality of its

training and the 2020 - 2025 interval is the best that fits because if it's too old it can conclude some different values than today's normal prices.

6. Conclusion

In conclusion of all those experiments and combinations of everything we could see that the best combination of each is while using LSTM with 2 hidden layers first one is 128 neurons and the second one with 64 neurons while checking the last 30 days and guessing the next 10 days with the information that it has and using 200 epochs while looking between 2020 - 2025. Everything we choose should be tested with different combinations and it shouldn't be too much or less.

7. Limitations and Future Recommendations

One of the biggest limitations was time and knowledge. We need to learn a lot to be better in AI coding and give much more time and effort in it so we can make a really high-quality product which would be more well designed and without any lack of features. We need to continue learning AI so we can produce better products in the future. We couldn't add any stock news reader bot which is also a big topic of the stock market. We need to learn how to create and use an LLM model to read news so it would be more accurate. Also we need more time to make it the most accurate possible.

8. References

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